

UPSC Civil Services (Main) Simulation GS-I

DETAILED SOLUTIONS / ANSWER KEY | Time: 3 Hours | Maximum Marks: 250

Q1. Explain how carbonation and groundwater routing create a karst landscape.

[10 marks | 150 words]

Model answer (119 words; practice model, not official): Chemistry supplies dissolution while structure, recharge, gradient, base level and time organise surface and underground forms. Karst is soluble-rock terrain organised by dissolution and underground drainage; a cave is a natural void, while cavern usually describes a large chamber rather than a separate universal origin. Soil carbon dioxide dissolved in water forms weak carbonic acid that converts calcium carbonate to dissolved calcium and bicarbonate; degassing can reverse the balance and precipitate calcite. Effective karstification requires soluble and sufficiently thick rock, joints or bedding pathways, recharge, hydraulic gradient, outlet or base-level relation and adequate time. Epikarst stores and routes recharge near the surface, vadose water descends through air-filled openings, and phreatic passages develop below the water table under saturated flow.

Q2. Explain lake-level change through a water-budget framework.

[10 marks | 150 words]

Model answer (91 words; practice model, not official): Storage responds to multiple inflows and losses, so attribution requires scale, period and connectivity rather than one visible driver. Lake storage changes with precipitation, inflow and groundwater inputs minus evaporation, outflow and seepage; level change cannot be assigned to one driver without a budget. Depth, area, residence time, temperature, wind and inflow density influence stratification and turnover; tropical and shallow lakes need not follow a simple temperate seasonal model. Diversion, damming, groundwater extraction, catchment change and warming-driven evaporation can shrink lakes; Aral Sea and Lake Chad illustrate different combinations and trajectories.

Q3. Explain why the world distribution of mineral and energy resources is better treated as a deduction from Earth history than as a list of producing countries.

[10 marks | 150 words]

Model answer (112 words; practice model, not official): Distribution follows geological history because shields concentrate metallic ores, sedimentary basins hold fuels and lateritic weathering profiles concentrate bauxite, while accessibility and economics decide which occurrences become usable resources. Mineral and energy resources are unevenly distributed because occurrence depends on geological history, rock structure, sedimentary basins, tectonics and climate-driven surface processes. Shield and igneous-metamorphic terrain is associated with metallic ores, while sedimentary basins hold coal, petroleum and natural gas. Tropical weathering and lateritisation concentrate bauxite on suitable plateaux and hilltops, so the bauxite map does not simply repeat the shield-ore map. Geology fixes occurrence only; grade, depth, transport distance, technology and price decide whether an occurrence becomes a usable resource at all.

Q4. Explain the tribe-caste continuum and state the limits of the thesis.

[10 marks | 150 words]

Model answer (106 words; practice model, not official): The continuum usefully describes long interaction between some communities and caste society, but it cannot dissolve distinct self-identification or the constitutional Scheduled Tribe category, so it is an analytical device with a legal boundary. Tribe has no single universal sociological checklist, while a Scheduled Tribe is a constitutionally notified category under Article 342, so older anthropological traits must never be used as a test of authenticity. The tribe-caste continuum is a historical sociological thesis that some communities long interacted with caste society through trade, migration or religious exchange and may show mixed features; it does not collapse tribal identities into caste or alter Scheduled Tribe status.

Q5. Distinguish endogamy from exogamy and explain why both rules can operate in the same marriage.

[10 marks | 150 words]

Model answer (111 words; practice model, not official): Endogamy fixes the outer boundary of the marriage circle while exogamy excludes the clan lineage inside it, so the two rules are complementary levels of one system rather than competing norms. Caste endogamy restricts marriage to within the caste while gotra exogamy restricts it to outside the clan lineage within that caste, so the two rules operate at different levels and typically apply together. Indian marriage forms include monogamy, regionally rare polygamy, endogamy, exogamy, arranged marriage and a rising share of self-choice marriage whose spread varies sharply by region, class,

education and generation. Patrilineal systems trace descent, surname and property transmission through the father's line and constitute India's dominant kinship pattern.

Q6. Distinguish Swadeshi from boycott in the anti-partition movement.

[10 marks | 150 words]

Model answer (115 words; practice model, not official): Boycott withdrew consumption and cooperation, while Swadeshi built indigenous production, education and self-reliance; the two were negative and constructive arms of one strategy. A mass meeting at Calcutta Town Hall adopted the Boycott Resolution on 7 August 1905, a key launch point of the Swadeshi movement. Swadeshi promoted indigenous goods, enterprise and self-reliance as both an economic programme and a political discipline. Boycott rejected foreign goods and, in more militant programmes, extended toward official schools, courts and functions where feasible. The National Council of Education was founded in 1906, and Bengal National College was associated with Aurobindo Ghosh. The militant programme used passive resistance, self-help and withdrawal of cooperation to reduce dependence on colonial institutions.

Q7. Distinguish the boycott programme from the constructive programme of Non-Cooperation.

[10 marks | 150 words]

Model answer (102 words; practice model, not official): The boycott programme was visible and nationwide but reversible within two years, while the constructive programme of khadi, education and unity work built durable institutions and cadre that outlasted the movement's withdrawal. The programme escalated through renunciation of titles, boycott of schools, courts and councils, economic boycott of foreign cloth, constructive work, and a final planned stage of mass civil disobedience and tax refusal. The boycott phase produced durable national institutions including Jamia Millia Islamia, Kashi Vidyapith, Gujarat Vidyapith and Bihar Vidyapith. The constructive programme of khadi, charkha, swadeshi, national education, panchayats and Hindu-Muslim unity built an alternative moral economy alongside boycott.

Q8. Explain why Indian tribal policy after 1947 rejected both isolation and assimilation, and state what it put in their place.

[10 marks | 150 words]

Model answer (111 words; practice model, not official): Both alternatives were rejected for stated reasons - isolation was insulting and by then impossible, assimilation would destroy identity and expose land and forest to outsiders - and the replacement was integration on two parameters that had to be held together. Verrier Elwin, who lived nearly all his life among the tribal people of central and north-eastern India, was one of the formative influences on the new government's policies and recorded that under British rule tribal people suffered oppression and exploitation as merchants and liquor-vendors cajoled, tricked and swindled them until their broad acres dwindled; his influence underwrote a policy of integration without imposition rather than a policy of rescue or of tutelage.

Q9. Explain why the Qutb complex is both a powerful and a limited source for understanding early Indo-Islamic architecture.

[10 marks | 150 words]

Model answer (130 words; practice model, not official): The Qutb complex is powerful evidence because several processes remain visible in one site. The Qutb Minar itself is layered: Aibak began it, Iltutmish substantially completed it, and Firoz Shah later repaired upper work. The site also preserves Quwwat-ul-Islam mosque, whose inscription records reuse of materials from twenty-seven Hindu and Jaina temples. Alai Darwaza then demonstrates a mature scientific handling of arch and dome. Thus conquest, reuse and engineering confidence can all be taught from one monument cluster. Yet the same site is also a limit. Quwwat-ul-Islam gives a monument-specific spolia case; it cannot be stretched into a universal claim about every Sultanate building. Nor should the entire minar be credited to one ruler. So the Qutb complex is valuable precisely when its evidence is used with chronology and caution.

Q10. Why did England industrialise first?

[10 marks | 150 words]

Model answer (109 words; practice model, not official): England's lead arose from a conjunction of capital, coal, labour release, agricultural surplus, markets, transport and commercial institutions rather than a single invention. Expanding trade, colonies and accumulated wealth supplied investible capital, but colonial gains were one element rather than the sole cause. England's abundant coal and iron supported energy-intensive machinery, though resources required markets, institutions and capital to become productive. The enclosure movement released landless labour for factories, linking agricultural change to industrial wage dependence. Crop rotation, manuring and new implements raised food supply and helped sustain a non-farming urban population. Overseas shipping and markets encouraged scale, while a commercially favourable post-seventeenth-century political-legal order reduced friction for enterprise.

Q11. Compare environmental hazards and human adaptations in hot and mid-latitude deserts.

[15 marks | 250 words]

Model answer (169 words; practice model, not official): Thermal regimes differ, but erratic storms, groundwater dependence and mobile or engineered water strategies organise both. Source notes commonly use annual precipitation below about 25 centimetres for true-desert examples, but the defining condition is long-term water deficit and high variability, not a universal cutoff. Hot deserts are especially known for large diurnal ranges under clear skies, whereas mid-latitude deserts add very cold winters and a much larger annual range. Rare intense storms can exceed infiltration, producing flash floods in normally dry channels or wadis; low annual rainfall does not mean low flood hazard. Reduced or waxy leaves, thorns, water-storing tissues, deep or spreading roots and widely spaced growth reduce water loss or improve access; no single trait defines every desert plant. An oasis forms where groundwater, springs or river-fed irrigation make water accessible, permitting date palms and cultivation; it is not a rainfall-fed forest island. Pastoral mobility, wells, tanks, gravity conduits, exotic rivers, canals and modern irrigation organise desert settlement, while over-abstraction, waterlogging and salinity can relocate the constraint.

Q12. Assess India's Bengal-Brahmaputra region as a China-type climatic analogue.

[15 marks | 250 words]

Model answer (185 words; practice model, not official): The humid eastern margin of India shares the China type's hot wet summers and mild winters supporting rice and tea, but is governed by monsoon seasonality and displays the flood-fertility paradox of the Brahmaputra valley. India's humid eastern margin, the Bengal-Brahmaputra plains and the North-East, is the subcontinental counterpart of the warm temperate eastern margin China type climate, with hot wet summers with heavy monsoon rain and mild winters supporting rice and tea. Khullar's Bengal and Assam Plain region covers the entire plain of West Bengal and the Brahmaputra valley of Assam, receiving about 150 centimetres annual rainfall from the Bay of Bengal branch of the south-west monsoon, with climate classed hot sub-humid to humid to perhumid. Rich alluvial soils combined with heavy monsoon rainfall form a solid base for rice cultivation in the Bengal-Assam plain; the same moisture regime that sustains rice and tea also makes Assam India's most flood-prone region. The Brahmaputra valley's flood-fertility paradox means the same monsoon regime that sustains rice and tea also makes Assam India's most flood-prone region through intense rain, sediment-rich braided channels, bank erosion and floodplain occupation.

Q13. How far is efficient and affordable urban mass transport a key to the economic development of a city region?

[15 marks | 250 words]

Model answer (173 words; practice model, not official): Mass transport raises effective accessibility and enlarges the labour and consumption market of a city region, but its developmental payoff depends on network reach, land-use integration and complementary local capacity rather than on the line alone. Transport is the circulatory system of an economy because it moves raw materials, labour, finished goods, information and strategic power across space, and trade geography grows out of connectivity, cost, time and chokepoints. Modern freight is a chain of road collection, rail or water trunk haul and road distribution, so container handling, multimodal terminals and last-mile connectivity determine actual logistics cost more than headline modal tariffs. Transport lowers input and output costs and widens markets, while development generates the traffic that justifies investment, so network density mirrors economic geography as much as it creates it. A corridor can pass through a region without benefiting it when there are no junctions, no local loading and no complementary investment, and better connectivity can also expose weak local producers to outside competition and accelerate out-migration of the young and skilled.

Q14. Examine how cross-cutting identities help India manage cultural diversity without permanent conflict.

[15 marks | 250 words]

Model answer (138 words; practice model, not official): Cross-cutting cleavages prevent any single identity from predicting every disadvantage, and integrative civic and syncretic institutions convert difference into routine interaction rather than standing grievance. Where identity axes do not align, disadvantage on one axis is often offset on another, so no single identity always predicts deprivation and group conflict is less intense. Sufi dargahs, Bhakti saint-poets, shared local deities and regional festivals such as Onam, Bihu and Pongal create composite belonging that crosses religion and region. Common schools, markets, elections and public transport work as routine integrative spaces in which diverse groups interact, and internal migration for work normalises mixed neighbourhoods. Indian diversity runs along ethnic or ethno-linguistic, linguistic, religious, regional and caste-tribe axes that are layered and overlapping rather than parallel, so cohesion depends on whether the axes reinforce each other or cut across each other.

Q15. Assess the dominant caste as a tool for analysing local power in India.

[15 marks | 250 words]

Model answer (149 words; practice model, not official): The dominant caste explains everyday authority better than ritual rank because it fuses numbers, land and office, but reserved representation and market differentiation now contest it without eliminating it. A dominant caste is numerically strong, owns a significant share of local land and wields political influence in a locality, whether or not it ranks highest ritually, so dominance is a better predictor of everyday local power than ritual rank. Srinivas derived both Sanskritisation and the dominant caste from fieldwork in Rampura village, Karnataka, which supplies the named-scholar spine every caste-mobility answer needs. Overlap describes localities where caste rank and economic class coincide, while cross-cutting

describes class differentiation inside a single caste, which is increasingly common with urban and market employment. Jati-level associations and caste panchayats arbitrate disputes, enforce endogamy norms and organise political mobilisation, which is how caste re-enters public life as organisation and network rather than as ritual.

Q16. Examine the regional diversity of early revolutionary nationalism.

[15 marks | 250 words]

Model answer (199 words; practice model, not official): Bengal cells, Maharashtra's Abhinav Bharat, north Indian networks and overseas centres shared armed anti-imperialism but differed in organisation, constituency and tactical opportunity. In 1897 the Chapekar brothers killed plague commissioner W.C. Rand and Lieutenant Ayerst in Poona, an armed precursor to the later phase. V.D. Savarkar's Mitra Mela developed into Abhinav Bharat in 1904, linking oath-bound organisation, political education and armed action. Anushilan Samiti and Dacca Anushilan were related but distinct Bengal networks; Barindra Ghosh's circle and Jugantar formed another centre. Jugantar was both a revolutionary network and a newspaper launched in 1906; Sandhya and Bande Mataram belonged to the wider militant press. In December 1909 Anant Kanhere killed Collector A.M.T. Jackson at Nasik; the prosecution exposed links associated with Abhinav Bharat. On 23 December 1912 a bomb was thrown at Viceroy Hardinge in Delhi; Rash Behari Bose and Sachin Sanyal were linked to the wider network. Before Ghadar, Tarak Nath Das published Free Hindustan and G.D. Kumar published Swadesh Sevak among Indian migrants on the Pacific coast. The Hindi Association of the Pacific Coast was organised in May 1913 with Sohan Singh Bhakna as president and Lala Har Dayal as a key leader.

Q17. Champaran, Ahmedabad and Kheda were laboratories of satyagraha rather than repetitions of one formula. Discuss.

[15 marks | 250 words]

Model answer (203 words; practice model, not official): The three campaigns shared disciplined non-violence and fact-based negotiation but differed in constituency, grievance, opponent, tactic and settlement, allowing Gandhi to test a transferable repertoire. Satyagraha joined truth, non-violence, voluntary self-suffering, openness and disciplined refusal of an unjust demand. Raj Kumar Shukla persistently brought Gandhi to Champaran in 1917, showing that local agency preceded national intervention. Champaran cultivators challenged tinkathia, which required indigo cultivation on 3/20 of a tenant's holding, and planter demands associated with release from that system. Gandhi's team recorded about 8,000 cultivator statements; an official inquiry led to a 25% refund of illegal dues, while later abolition of tinkathia must be kept distinct from that refund. In the 1918 Ahmedabad plague-bonus dispute, workers initially sought 50%, owners offered 20%, and Gandhi supported a 35% wage increase. After a twenty-one-day strike, Gandhi's three-day fast, his first in an Indian struggle, reinforced the pledge and pressed the dispute toward arbitration; sources differ on the final percentage outcome. The 1918 Kheda campaign sought suspension of land revenue where assessed produce was four annas or less, one-quarter of normal, under the government's own rules. Vallabhbhai Patel and Indulal Yagnik helped organise Kheda, showing that Gandhian action depended on local and regional leadership.

Q18. Trace the process by which accession in 1947 was converted into administrative and constitutional integration.

[15 marks | 250 words]

Model answer (196 words; practice model, not official): Integration was a sequence and not an event: accession on three subjects, then merger and the creation of unions, then democratisation inside the former states, and finally constitutional incorporation that dissolved princely subjecthood into citizenship. The second and harder stage of full integration began in December 1947, when smaller states were merged with neighbouring provinces or grouped into new unions, so accession was only the first of two distinct achievements and never the whole of integration. Large numbers of small states were consolidated into new unions such as Madhya Bharat, Rajasthan, the Patiala and East Punjab States Union, Saurashtra and Travancore-Cochin, while Mysore, Hyderabad and Jammu and Kashmir retained their original form as separate states of the Union. Rulers who surrendered power received privy purses and retained titles, succession and ceremonial privileges, and these settlements were not abolished during the initial integration; these privileges were abolished in 1971. The Constituent Assembly, presided over by Dr Rajendra Prasad with B.R. Ambedkar chairing the Drafting Committee, adopted the Constitution on 26 November 1949 and it came into force on 26 January 1950, which is the date on which India became a republic and princely subjects became citizens.

Q19. "Though the great Cholas are no more yet their name is still remembered with great pride because of their highest achievements in the domain of art and architecture." Comment.

[15 marks | 250 words]

Model answer (233 words; practice model, not official): The statement is justified because Chola achievement in art and architecture was not a single monument but a mature civilisational package of temple building, sculpture, painting and institutional organisation. Architecturally, the sequence begins with early shrines such as Narttamalai and develops through increasingly complex mandapa and devakoshtha arrangements before reaching the imperial climax at Brihadishvara, Thanjavur. There the soaring vimana, granite construction, axial planning, Nandi mandapa, enclosure and mural programme combine scale with technical control. Rajendra I's Gangaikondacholapuram and the later Airavatesvara temple at Darasuram show that this tradition remained inventive, while the later phase marks the increasing prominence of the gopura. Chola memory also rests on sculpture and painting. Their bronze tradition,

above all Nataraja, became the classic image of kinetic divinity in Indian art. Brihadishvara's murals and Chola wall-painting remains show that sacred narrative, devotion and royal presence were also rendered pictorially. Inscriptions deepen the legacy further: they document donations, festivals, daily worship and administration, proving that the temple was a living institution, not an inert monument. A balanced answer should add two cautions. First, not every Chola project survives equally well - Gangaikondacholapuram was left incomplete in comparison with Thanjavur. Second, the achievement depended on large political and material resources that the stone record does not fully explain. Even with these limits, the Cholas are remembered with pride because they fused scale, aesthetics, ritual and administration into one durable artistic order.

Q20. German unification was a revolution from above. Discuss.

[15 marks | 250 words]

Model answer (153 words; practice model, not official): Popular nationalism created legitimacy, but Prussian institutions, Bismarck's diplomacy and three wars determined timing, boundaries and the authoritarian form of unity. The revolutions of 1848 put national unity on the agenda but failed, redirecting both movements from liberal idealism toward statecraft, diplomacy and war. Otto von Bismarck became Prussia's minister-president in 1862 and used Realpolitik, practical state interest rather than abstract principle, to direct unification. Prussia and Austria defeated Denmark in 1864 over Schleswig-Holstein, opening Bismarck's three-war sequence. Prussia defeated Austria in the Seven Weeks' War of 1866, excluded Austria from German affairs and established the North German Confederation. War against France in 1870-71 rallied the south German states to Prussia and completed the military-diplomatic unification sequence. The German Empire was proclaimed at Versailles in January 1871 under William I of Prussia. The new Empire combined a parliament and federal institutions with Prussian dominance and monarchical control of the army and foreign policy.

END OF DETAILED SOLUTIONS